

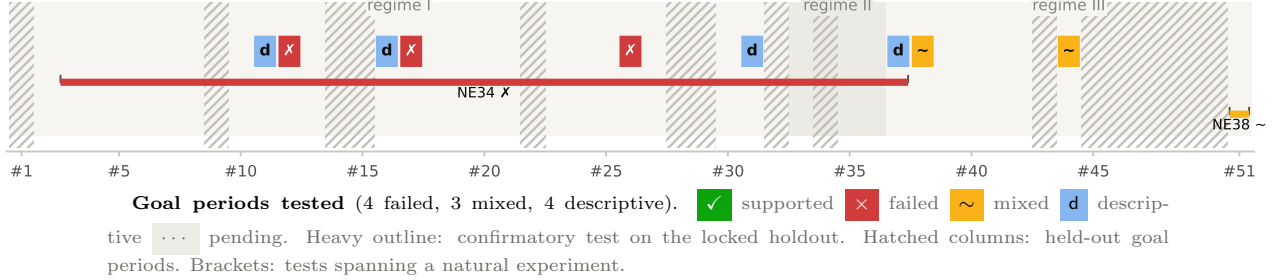
H10: Goals are Legendre pushes: a swarm's response to an assigned goal is predictable from its unforced fluctuations

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. In the AI Village, goals are quenches toward a common target, not Legendre pushes: in both embedding models free-week fluctuations do not predict who follows a goal, and a reassigned agent lands where its unforced content never went.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Can a swarm's unforced fluctuations predict how it lines up behind an assigned goal? Agents' 30-min statement embeddings were projected on the goal direction $\hat{g}$. In three free $\rightarrow$ assigned pairs (#11 $\rightarrow$ #12, #16 $\rightarrow$ #17, #37 $\rightarrow$ #38) one field λ was fitted to the mean shift and the rest predicted. Alignment rose 1.4–3.1 free-week SDs, outside linear response. **The core test failed:** fluctuations did not predict who moved ($r = -0.37, -0.69, +0.20$; Stouffer $p = 0.89$). **Round 1b** (shared goal vectors, a second embedding model, dedupe, style residualization): failed in all 8 input configurations. Variance along $\hat{g}$ grew while transverse variance held. New native tests: one agent's reassigned goal moved it 10–20 SDs, beyond anything its unforced content visited; an agent-set goal moved agents but again not in fluctuation order. Holdout frozen, not run.

PHYSICAL MODEL

Model 11, mean-field $O(n)$ in $n = 32$ whitened embedding dimensions. $\mathbf{s}_i$: agent i 's unit statement vectors; $y = \hat{\mathbf{z}} \cdot \hat{\mathbf{g}}$, alignment with the goal field direction; x_i , its 30-min window mean, with free-week mean μ_i , noise-corrected variance $\kappa_{2,i}$, third cumulant $\kappa_{3,i}$. The goal is a linear field $\lambda = \beta h$ along $\hat{g}$; K is the cumulant generating function; J_0 the mean-field coupling; g the loop gain; C_F the free-week window covariance.

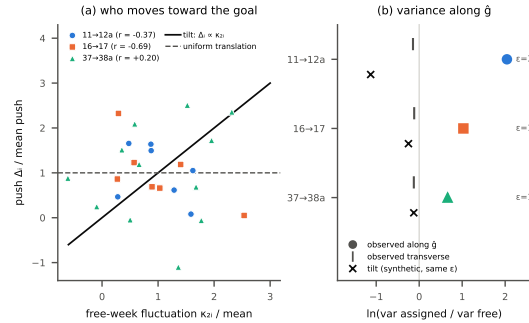
$$P_A(\mathbf{s}) \propto P_F(\mathbf{s}) e^{\lambda \sum_i \hat{\mathbf{g}} \cdot \mathbf{s}_i}, \quad G = F - \lambda m$$

$$K_A(t) = K_F(t + \lambda) - K_F(\lambda) \Rightarrow \kappa_n^A = K_F^{(n)}(\lambda)$$

$$\Delta \mu_i = \lambda \kappa_{2,i} + \frac{\lambda^2}{2} \kappa_{3,i}, \quad \kappa_{2,i}^A = \kappa_{2,i} + \lambda \kappa_{3,i}$$

$$\chi_{\text{swarm}} = \chi_0 / (1 - \beta J_0 \chi_0), \quad g = \beta J_0 \chi_0 = 1 - V_{\text{indep}} / V$$

$$\Delta \mu = \lambda C_F \hat{g}$$



Observables. (a) Each agent's push toward the assigned goal vs. its free-week fluctuation along $\hat{g}$, both divided by the pair mean. The tilt predicts the solid line, uniform translation the dashed one; the regime-I pairs slope the wrong way. (b) Change in variance along $\hat{g}$ (colored), along transverse directions (bars), and for an exact tilt of a bounded mean-field swarm at the same push ($\times$, synthetic). The goal disperses the swarm along $\hat{g}$ instead of narrowing it.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a negative result, useful only in a narrow sense. The appealing promise, reading how steerable a swarm is from how it fluctuates when nobody steers it, does not hold here. Goal changes are huge kicks (median about 2.5 SDs), not small fields, and the agents that wander most are not the ones that follow most. What an operator learns: an assigned goal acts like a target everyone converges on and a new mode the swarm toggles collectively, not a gentle tilt of existing preferences. So pre-goal behavior is a poor forecast of compliance. The instrument (goal-direction projection, noise-corrected variance) is cheap on any agent log, but the fluctuation–response physics it was built to test is, in this village, a curiosity.

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RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b (bge / gte)	Verdict
P1 fluctuations order who moves	r $-0.37, -0.69, +0.20 \rightarrow$ same / $\times$ $-0.53, -0.49, -0.09$; 8/8 configs fail	$\times$
Push $\Delta\bar{\mu} > 0$	3/3 pairs; 19/19 kickoffs $\rightarrow$ same $\checkmark$ in all configs	$\checkmark$
P2 variance by the tilt	untestable ($\varepsilon > 0.34$); gte n/a #37→#38 would pass (ε 0.69)	n/a
Cross-kickoff man > 0	Spear- $-0.56 \rightarrow -0.56 / -0.59$	$\times$
G44 assigned vs own-choice room	#best ε 0.97 / 1.55; #rest none; $\sim$ gap +0.17	$\sim$
NE38 one agent re-signed	ε 10 / 20; post mean beyond the pre support $\sim$ (tilt $\times$)	$\times$
G26 agent-set goal	push $\varepsilon \approx 1$; P1 r $-0.14 / -0.63$	$\times$

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	0	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

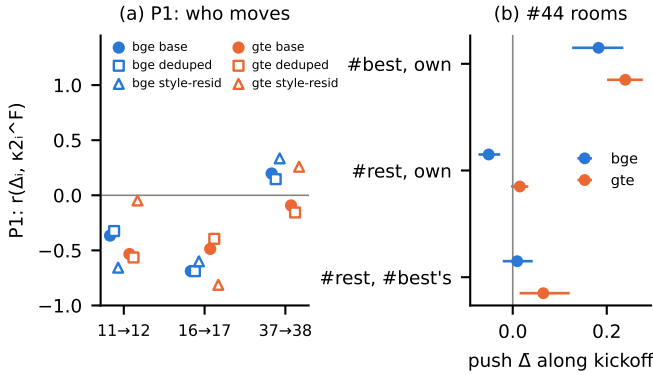
CAVEATS

$\hat{g}$ is a text-embedding proxy for the field; gte shrinks the regime-III push and makes its variance tilt-compatible, so regime III is model-dependent. $N = 7$ in regime I. NE38's concurrent control is one incumbent, which drifted 1–3 SDs along its own goal. The #44 assigned room has 4 agents and three days. Natives are exploratory; #22→#23 and #31→#32 remain blind.

ROUND 2

Re-map goals as a two-state (on/off-goal) occupancy and test field vs contagion onsets (H10-R1/R2); run `confirm.py` only after Vivian's sign-off.

- **Essence.** How does a new goal propagate: as a field acting on everyone at once, or as contagion from early adopters?
- **H10-R1** Adoption onsets after a kickoff: a field predicts simultaneous onsets; contagion predicts onsets ordered by exposure to early adopters.
- **H10-R2** Goals act on a two-state on-goal/off-goal occupancy (HH130).
- **H10-R3** During the week, goal-specific content relaxes back toward family-prior directions (E4).



Second figure. (a) P1 per pair under every round-1b input (blue bge, orange gte; filled base, squares deduped, triangles style-residualized): the regime-I pairs stay negative. (b) #44 native: the assigned room moves along its own kickoff, the self-chosen room does not (90% CIs).

GOAL PERIODS IN DETAIL

G11 d 1b: descriptive (both models)
 G12 $\times$ 1b: failed (both models)
 G16 d 1b: descriptive (both models)
 G17 $\times$ 1b: failed (both models)
 G26 $\times$ failed (round 1b native; both models)
 G31 d 1b: descriptive (both models)
 G37 d 1b: descriptive (both models)
 G38 $\sim$ 1b: mixed (bge mixed, gte failed)
 G44 $\sim$ mixed (round 1b native; model-dependent)
 NE34 $\times$ 1b: failed (both models)
 +1 more periods: see the card.

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